



BITS Pilani
Hyderabad Campus

Birla Institute of Technology & Science, Pilani
Hyderabad Campus
Second Semester 2024 – 2025
End-Semester (Compre) Examination

Course No. CS F407

Course Name. Artificial Intelligence

Your ID No: _____

Name: _____

Date: 2nd May, 2025

Total Marks: 90

Time: 90 Mins

General Instructions:

- This Question Paper contains two parts: **Part-(A)** and **Part-(B)**. Answer the questions of Part-(A) in a separately dedicated Answer Sheet. (Only ONE correct answer per question)
- **Separate your Answer Sheet for Part-(A) and submit it after 1 hour.**
- Answers to all questions of **Part-(B)** must be written in the answer booklet provided and **in serial order with all sub-parts of each question together.**
- Only answers written in ink (**NOT in pencil**) will be evaluated.
- **The Answer Booklet is for Part-(B).**
- All necessary working must be shown in the answer sheet itself. Merely writing the answers is not sufficient.
- Write your Name and ID No in all pages of Answer Sheets / Booklets. **(Best wishes)!!**

Part - (A)

1. In probabilistic hypothesis for supervised learning, consider a multi-class classification problem with k classes. The hypothesis function typically takes the form of a multinomial distribution: $h_{\theta}^{(j)}(x) = p(y = j | x; \theta)$ for each class $j \in \{1, 2, \dots, k\}$. Which of the following correctly describes how these hypothesis are structured? **(2 Marks)**
- A. Each class shares identical parameters θ , ensuring uniform prediction capability across all classes.
- B. Parameters θ are arbitrarily chosen and do not represent distinct probability distributions for each class.
- C. The hypothesis function for each class has its own distinct parameter set, representing separate probability distributions.
- D. The probability of class membership is independent of the input data x .

Solution + Explanation: In a multinomial classification problem (e.g., softmax regression), each class j has its own parameters with a unique conditional probability distribution $p(y = j | x; \theta_j)$. Hence, option **(C)**

2. In a calibrated pinhole camera model with intrinsic matrix K given by:

$$K = \begin{bmatrix} 1200 & 0 & 640 \\ 0 & 1200 & 480 \\ 0 & 0 & 1 \end{bmatrix}$$

a 3D point at $(X, Y, Z) = (120, 90, 600)$ is projected onto the 2D image plane. Determine the coordinates of the projected point. **(2 Marks)**

- A. (880, 660) B. (880, 6600) C. (780, 580) D. (800, 700)

Solution and Explanation:

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \frac{1}{Z} \cdot K \cdot \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \frac{1}{600} \cdot \begin{bmatrix} 1200 & 0 & 640 \\ 0 & 1200 & 480 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 120 \\ 90 \\ 600 \end{bmatrix} = (880, 660, 1)^T$$

Hence, correct option **(A)**

3. A grayscale image of dimension 512×512 is convolved with a kernel of dimension 5×5 using zero-padding and stride of 2. The dimension of the convolved image is: **(2 Marks)**

- A. 254×254 B. 256×256 C. 128×128 D. 508×508

Solution and Explanation:

$$\text{The output size will be} = \left\lfloor \frac{W - K + 2P}{S} \right\rfloor + 1 = \left\lfloor \frac{512 - 5}{2} \right\rfloor + 1 = \left\lfloor \frac{507}{2} \right\rfloor + 1 = 254$$

Hence, option **(A)**

4. Suppose for linear regression task, the cost function is $J(\theta) = \frac{1}{2m} \sum_{i=1}^m (\theta^T x^{(i)} - y^{(i)})^2$. Then what is the correct expression for the gradient $\nabla J(\theta)$ in terms of the design (or data) matrix X and labels y ? **(2 Marks)**

- A. $\frac{1}{m}(X\theta - y)$ B. $\frac{1}{m}X^T(X\theta - y)$ C. $X^T(y - X\theta)$ D. $\frac{1}{m}(y - X\theta)$

Solution and Explanation: Cost function: $J(\theta) = \frac{1}{2m} \sum_{i=1}^m (\theta^T x^{(i)} - y^{(i)})^2$, Gradient: $\nabla J(\theta) = \frac{1}{2m} X^T(X\theta - y)$, and therefore $\frac{1}{m} X^T(X\theta - y)$ Hence, option **(B)**

5. Which is logically equivalent to $\neg \exists x (Student(x) \wedge Passed(x))$ **(2 Marks)**

- (A) $\exists x \neg (Student(x) \wedge Passed(x))$
 (B) $\forall x (Student(x) \rightarrow \neg Passed(x))$
 (C) $\forall x \neg (Student(x)) \vee Passed(x)$
 (D) $\forall x (Student(x) \wedge \neg Passed(x))$

Solution and Explanation: $\neg \exists x (Student(x) \wedge Passed(x)) \equiv \forall x (Student(x) \rightarrow \neg Passed(x))$, using $\neg \exists x \phi(x) \equiv \forall x \neg \phi(x)$, so the answer is $\forall x, (Student(x) \rightarrow \neg Passed(x))$. Hence, correct option **(B)**

6. Suppose we train a linear regression model $\hat{f}(x) = \theta^T x$ by minimizing Mean Squared Error on the training set $X = \{.., (x, y), ..\}$. Let **the true underlying** function be $f(x) = \mathbb{E}[y|x]$, and the total expected prediction error at a new point x_0 is $\mathbb{E}[(\hat{f}(x_0) - f(x_0))^2]$. Which of the following decompositions correctly represents the expected error? **(2 Marks)**

- A. Variance of noise + (Bias)² only

- B. $(\text{Bias})^2 + \text{Variance} + \text{Irreducible error}$
- C. $(\text{Bias})^2 + \text{Model Complexity}$
- D. $\text{Variance} + \text{Model Overfitting Error}$

Solution and Explanation: The correct option (B)

7. Consider a second-order Markov model defined as: $P(X_t|X_{t-1}, X_{t-2}, \dots) = P(X_t|X_{t-1}, X_{t-2})$. Now, we train a first-order Markov model on a process that actually follow this second-order process. Which of the following statements is the most accurate? (2 Marks)
- A. The model will overfit to recent transitions and perform poorly on long-term prediction.
 - B. The model will assign incorrect transition probabilities due to incorrect independence assumptions.
 - C. The model will perform better due to reduced variance in parameter estimation.
 - D. The model will behave identically as long as the state space is fully observable.

Solution and Explanation: The correct option (B)

8. Let $\phi = \forall x (P(x) \rightarrow Q(x))$. Which statement is logically equivalent to ϕ ? (2 Marks)
- A. $\phi \models \forall x (\neg Q(x) \rightarrow \neg P(x))$
 - B. $\forall x (\neg Q(x) \rightarrow \neg P(x)) \models \phi$
 - C. $\phi \equiv \forall x (\neg Q(x) \rightarrow \neg P(x))$
 - D. They are logically unrelated

Solution and Explanation: Given: $\phi = \forall x (P(x) \rightarrow Q(x))$, by **contrapositive transformation**, this is equivalent to: $\forall x (\neg Q(x) \rightarrow \neg P(x))$, so the correct answer is: $\forall x (\neg Q(x) \rightarrow \neg P(x))$. Hence, correct option (C)

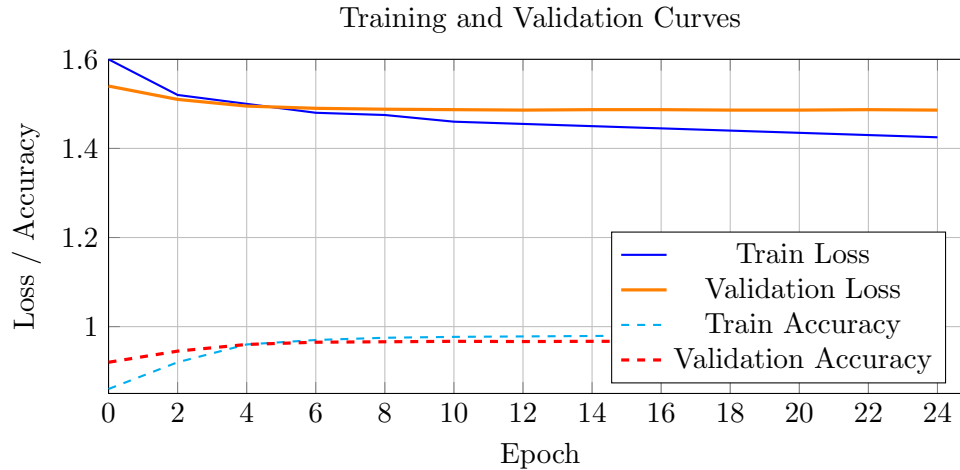
9. Consider a Simulated Annealing (SA) setup where the initial temperature $T_0 = 100$, the cooling schedule $T_{i+1} = 0.5 \times T_i$, and the iteration stops at $T < 1$. If the cost landscape contains multiple local minima, why might this annealing setup fail to reach a global optimum? (2 Marks)
- (A) Geometric cooling converges too slowly
 - (B) No worse states accepted below $T = 10$
 - (C) Cooling too fast, losing randomness early
 - (D) Exponentials aren't suitable for probabilities

Solution and Explanation: Given: $T_0 = 100$, $T_{i+1} = 0.5 \times T_i$, stop when $T < 1$, with multiple local minima, and SA needs high T to explore; cooling too fast $\Rightarrow$ early freezing, loses randomness. Hence, correct option (C)

10. You are given a deep neural network with 4 layers (excluding input) and their sizes are Input: 128, Layer-1: 256, Layer-2: 128, Layer-3: 64, and Output: 10. Then, what is the correct shape of the weight matrix $\mathbf{W}^{[3]}$, assuming that each layer l uses the update $\mathbf{a}^{[l]} = \text{ReLU}(\mathbf{W}^{[l]}\mathbf{a}^{[l-1]} + \mathbf{b}^{[l]})$? (2 Marks)
- A. 64×128
 - B. 128×64
 - C. 64×10
 - D. 10×64

Solution and Explanation: Layer-2 has 128 units, Layer-3 has 64 units of neurons. The general form $a^{[l]} = \text{ReLU}(W^{[l]} \cdot a^{[l-1]} + b^{[l]})$. $W^{[l]}$ maps $a^{[l-1]} \rightarrow a^{[l]} \Rightarrow W^{[3]}$ shape: 64×128 . Hence, correct option **(A)**

11. Suppose that after 10 epochs, the validation loss plateaus while training loss continues to decrease as shown below. Then What does this most likely indicate? **(2 Marks)**



- A. The model is underfitting on both training and validation sets
- B. The model is generalizing better as the training loss decreases
- C. The model is overfitting the training data
- D. The model has perfectly fit the training and validation sets

Solution and Explanation: The correct option is **(C)**

12. A data scientist implements Expectation Maximization (EM) on a Gaussian mixture model (GMM) with two clusters but skips the normalization step in the E-step, using this as the assignment $z_{nk} = \pi_k \cdot \mathcal{N}(x_n | \mu_k, \Sigma_k)$ (**Note:** the variable notations are commonly used). If they proceed with the M-step using this faulty z_{nk} , what is the most likely consequence? **(2 Marks)**

- A. The assignments will still be valid as long as π_k sums to 1
- B. The updates will behave like hard k-means assignments
- C. The M-step will overweight denser Gaussian mixtures, corrupting convergence
- D. The updates will not change μ_k but still adjust Σ_k

Solution and Explanation: Since the The E-step is implemented without normalization, they will start favoring low or narrow variance Gaussians from the mixtures with likelihood peaks that are higher. In contrast, during the M-Step the cluster will be favored disproportionately. Hence, correct option **(C)**

13. Which of the following statements is true regarding the difference between k -Means and Expectation Maximization (EM) for Gaussian Mixtures? **(2 Marks)**

- A. EM uses hard assignments like k-means but updates covariances

- B. k-means uses posterior probabilities for assignment
- C. EM computes soft assignments $p(k|x_n)$, while k-means does not
- D. k-means supports probabilistic modeling of data distribution

Solution and Explanation: Correct option (C)

14. A hospital uses a hybrid Bayesian model where *BloodSugar* (continuous variable) level depends on *MedicationStatus* (discrete: On/Off) and BMI (continuous variable) using a linear Gaussian conditional: $BloodSugar \sim \mathcal{N}(a_m \cdot BMI + b_m, \sigma^2)$. Which of the following statements is true? **(2 Marks)**

- A. EM uses hard assignments like k-means but updates covariances
- B. Parameters a_m, b_m, σ_m change depending on whether medication is On or Off
- C. This is equivalent to fitting two independent Gaussian regressions
- D. The model must treat BMI as a categorical variable for inference

Solution and Explanation: Correct option (C)

15. An e-commerce platform uses a Bayesian Network to model ***WillBuy*** (binary variable) based on ***DiscountPercentage*** (continuous variable) and ***UserCategory*** (discrete: Regular, Prime). They decide to model the probability of ***WillBuy*** using a logistic (sigmoid) function on the normalized cost: $p(\mathbf{WillBuy} = 1 | \text{Price}) = \frac{1}{(1 + e^{(c - \mu)/\sigma})}$. Which of the following is the most accurate? **(2 Marks)**

- A. The model ensures Prime users always buy at lower prices
- B. μ sets a price threshold where buy probability is 0.5; σ controls how sharply the transition occurs
- C. μ is the average price across all user categories; σ is the price variance
- D. The sigmoid ensures buying probability is uniform across user types

Solution and Explanation: Correct option (B)

Part - (B)

1. Suppose a medical test returns two features (x_1, x_2) . Once such test data $\mathcal{D} = \{(x_1^i, x_2^i) \mid \forall i = 1 \dots 4\}$, a hypothesis $\mathcal{H}_\theta = \sigma(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$ is used for a binary label $y \in \{0, 1\}$ prediction tasks where σ is the 'SoftMax' function:

i	$x_1^{(i)}$	$x_2^{(i)}$	$y^{(i)}$
1	0.5	1.0	1
2	1.5	0.8	1
3	1.0	1.2	0
4	0.2	0.4	0

- (a) Suppose the co-efficient vector $\theta = (\theta_0, \theta_1, \theta_2)^T = (0, 1, -1)^T$, then compute $\mathcal{H}_\theta(x^i)$ for each $i \in \{1, 2, 3, 4\}$. **(4 marks)**
- (b) Compute the gradient $\nabla J(\theta)$ and derive all the steps of the gradient-descent update for first iteration using $\alpha = 0.1$. What will be the updated value of θ ? **(6 marks)**

Solution and Explanation: (a) Computing $\mathcal{H}_\theta(x^{(i)})$: Let $\theta = (0, 1, -1)^T$
 $\mathcal{H}_\theta(x) = \text{Softmax}([\theta^T x, 0]) = [\text{Sigmoid}(z), 1 - \text{Sigmoid}(z)]$, with $\text{Sigmoid}(z) = \frac{1}{1+e^{-z}}$, $z = \theta_0 + \theta_1 x_1 + \theta_2 x_2$

i	$x_1^{(i)}$	$x_2^{(i)}$	$z = \theta^T x$	$\text{Sigmoid}(z)$	$\mathcal{H}_\theta(x^{(i)})$
1	0.5	1.0	-0.5	0.3775	[0.3775, 0.6225]
2	1.5	0.8	0.7	0.6682	[0.6682, 0.3318]
3	1.0	1.2	-0.2	0.4502	[0.4502, 0.5498]
4	0.2	0.4	-0.2	0.4502	[0.4502, 0.5498]

(b) Compute Gradient $\nabla J(\theta)$ and Updated Parameters

First define the logistic loss gradient:

$$\nabla J(\theta) = \frac{1}{m} \sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)}) \cdot x^{(i)}, \quad \text{where } \hat{y}^{(i)} = \sigma(z^{(i)}), \quad x^{(i)} = [1, x_1^{(i)}, x_2^{(i)}]$$

Next, the Error Calculation as follows:

i	$y^{(i)}$	$\hat{y}^{(i)}$	$\hat{y}^{(i)} - y^{(i)}$
1	1	0.3775	-0.6225
2	1	0.6682	-0.3318
3	0	0.4502	0.4502
4	0	0.4502	0.4502

Next, Computing the gradient w.r.t the parameters

Bias θ_0 : $\frac{1}{4}(-0.6225 - 0.3318 + 0.4502 + 0.4502) = -0.013475$

Weight θ_1 : $\frac{1}{4}(-0.6225 \cdot 0.5 - 0.3318 \cdot 1.5 + 0.4502 \cdot 1.0 + 0.4502 \cdot 0.2) \approx -0.0733$

Weight θ_2 : $\frac{1}{4}(-0.6225 \cdot 1.0 - 0.3318 \cdot 0.8 + 0.4502 \cdot 1.2 + 0.4502 \cdot 0.4) \approx -0.0604$

Step 4: Gradient Descent Update ($\alpha = 0.1$)

Initial: $\theta = [0, 1, -1]$

Update: $\theta_{\text{new}} = \theta - \alpha \cdot \nabla J(\theta) = [0 + 0.00135, 1 + 0.00733, -1 + 0.00604]$

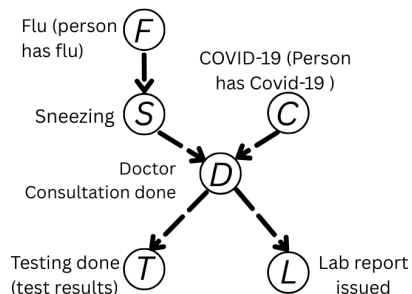
$\theta_{\text{new}} = [0.0013, 1.0073, -0.9940]$

Variable	Probability	S	C	$P(D = \text{true} S, C)$	$P(D = \text{false} S, C)$
$P(F = \text{true})$	0.05	true	true	0.95	0.05
$P(F = \text{false})$	0.95	true	false	0.90	0.10
$P(C = \text{true})$	0.02	false	true	0.60	0.40
$P(C = \text{false})$	0.98	false	false	0.05	0.95

F	$P(S = \text{true} F)$	$P(S = \text{false} F)$
true	0.80	0.20
false	0.10	0.90

D	$P(T = \text{true} D)$	$P(T = \text{false} D)$
true	0.92	0.08
false	0.10	0.90

D	$P(L = \text{true} D)$	$P(L = \text{false} D)$
true	0.88	0.12
false	0.05	0.95



2. Consider the previous Bayesian Network representing a simplified COVID-19 Diagnosis System (*on the right*) with the mentioned variables and conditional probability tables (*on*

the left).

Given that a Lab Report is issued ($L = \text{true}$) and Testing is done ($T = \text{true}$), calculate the probability that the person has Flu ($F = \text{true}$)? **(10 Marks)**

Solution and Explanation: Given $T = \text{true}$, $L = \text{true}$ We need to compute $P(F) \cdot P(S | F) \cdot P(C) \cdot P(D | S, C) \cdot P(T | D) \cdot P(L | D)$. Therefore, first need to compute the joint probability: $P(F, S, C, D, T, L) = P(F) \cdot P(S|F) \cdot P(C) \cdot P(D|S, C) \cdot P(T|D) \cdot P(L | D)$. Now, if we fix the $T = \text{True}$ and $L = \text{True}$, then the simple approach is to calculate the two joint probability distributions in the numerator and the denominator for all $2^3 = 8$ instances of S, C, D 's CPT table as follows (**Numerator terms when $F = \text{true}$**):

$$\sum_{S,C,D} P(F) \cdot P(S|F) \cdot P(C) \cdot P(D|S, C) \cdot P(T|D) \cdot P(L|D) = \boxed{0.02016}$$

$$\rightarrow S=\text{True}, C= \text{True}, D=\text{True} \text{ then } 0.05 \cdot 0.80 \cdot 0.02 \cdot 0.90 \cdot 0.92 \cdot 0.88 = \mathbf{0.0005806}$$

$$\rightarrow S=\text{True}, C= \text{True}, D=\text{False} \text{ then } 0.05 \cdot 0.80 \cdot 0.02 \cdot 0.10 \cdot 0.10 \cdot 0.05 = \mathbf{0.0000040}$$

$$\rightarrow S=\text{True}, C= \text{False}, D=\text{True} \text{ then } 0.05 \cdot 0.80 \cdot 0.98 \cdot 0.60 \cdot 0.92 \cdot 0.88 = \mathbf{0.0190423}$$

$$\rightarrow S=\text{True}, C= \text{False}, D=\text{False} \text{ then } 0.05 \cdot 0.80 \cdot 0.98 \cdot 0.40 \cdot 0.10 \cdot 0.05 = \mathbf{0.0007840}$$

$$\rightarrow S=\text{False}, C= \text{True}, D=\text{True} \text{ then } 0.05 \cdot 0.20 \cdot 0.02 \cdot 0.05 \cdot 0.92 \cdot 0.88 = \mathbf{0.0000081}$$

$$\rightarrow S=\text{False}, C= \text{True}, D=\text{False} \text{ then } 0.05 \cdot 0.20 \cdot 0.02 \cdot 0.95 \cdot 0.10 \cdot 0.05 = \mathbf{0.0000950}$$

$$\rightarrow S=\text{False}, C= \text{False}, D=\text{True} \text{ then } 0.05 \cdot 0.20 \cdot 0.98 \cdot 0.05 \cdot 0.92 \cdot 0.88 = \mathbf{0.0003986}$$

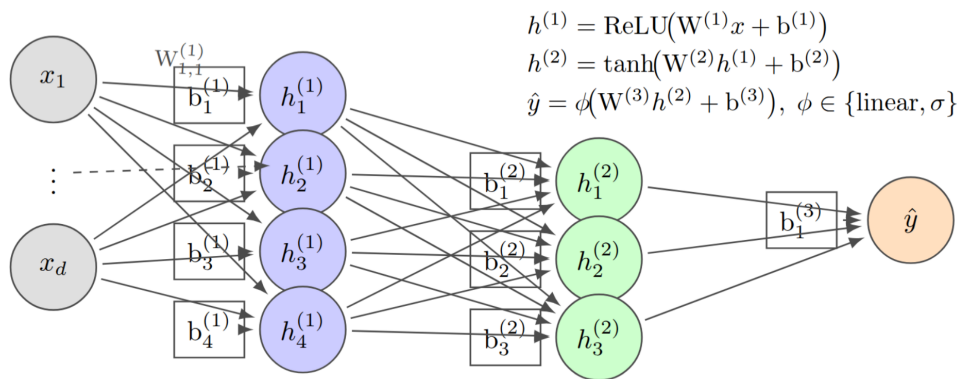
$$\rightarrow S=\text{False}, C= \text{False}, D=\text{False} \text{ then } 0.05 \cdot 0.20 \cdot 0.98 \cdot 0.95 \cdot 0.10 \cdot 0.05 = \mathbf{0.0007444}$$

and the Denominator term (for $F = \text{True}$ as well as $F = \text{False}$): Repeat the above computation and also for **F = false**. In the case of $F = \text{False}$, $P(F)$ will be = 0.95

$$\text{Denominator Sum} = \text{Numerator} + F=\text{false terms} = 0.02016 + 0.08546 = \boxed{0.10562}$$

The final value of the ratio will be $P(F = \text{true} | T = \text{true}, L = \text{true}) = \frac{0.02016}{0.10562} \approx \boxed{0.191}$

3. Use the below MLP for two tasks: (i) Binary classification using logistic regression, and (ii) Continuous value prediction using linear regression.



- (a) For a given input $\mathbf{x}$, derive the explicit expressions for $h^{(1)}$, $h^{(2)}$, and $\hat{y}$. **(3 Marks):**
(b) Compute the derivatives $\frac{\partial \mathcal{L}}{\partial W_{jk}^{(2)}}$ and $\frac{\partial \mathcal{L}}{\partial b_j^{(2)}}$. Explain the role of the derivative of the tanh and ReLU functions in your computation. **(7 Marks):**

Use consistent notation: Subscripts denote neuron indices, superscripts denote layer numbers (e.g., $W_{jk}^{(2)}$ connects the k -th neuron in layer 1 to the j -th neuron in layer 2).

Solution and Explanation:

- **First hidden layer:**

$$z^{(1)} = W^{(1)}\mathbf{x} + \mathbf{b}^{(1)} \in \mathbb{R}^4$$

$$h^{(1)} = \text{ReLU}(z^{(1)}), \quad h_j^{(1)} = \max(0, z_j^{(1)})$$

- **Second hidden layer:**

$$z^{(2)} = W^{(2)}h^{(1)} + \mathbf{b}^{(2)} \in \mathbb{R}^3$$

$$h^{(2)} = \tanh(z^{(2)}), \quad h_j^{(2)} = \tanh(z_j^{(2)})$$

- **Output:**

$$z^{(3)} = W^{(3)}h^{(2)} + b^{(3)}$$

$$\hat{y} = \phi(z^{(3)}), \quad \phi \in \{\text{linear}, \sigma\}, \quad \sigma(z) = \frac{1}{1+e^{-z}}$$

Therefore the explicit and expanded expression of the hidden layers and the output:

$$\mathbf{h}^{(1)} = \max \left\{ 0, \begin{bmatrix} W_{11}^{(1)} & W_{12}^{(1)} & \cdots & W_{1d}^{(1)} \\ W_{21}^{(1)} & W_{22}^{(1)} & \cdots & W_{2d}^{(1)} \\ W_{31}^{(1)} & W_{32}^{(1)} & \cdots & W_{3d}^{(1)} \\ W_{41}^{(1)} & W_{42}^{(1)} & \cdots & W_{4d}^{(1)} \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} + \begin{bmatrix} b_1^{(1)} \\ b_2^{(1)} \\ b_3^{(1)} \\ b_4^{(1)} \end{bmatrix} \right\}$$

then $\mathbf{h}^{(2)} =$

$$\tanh \left(\begin{bmatrix} W_{11}^{(2)} & W_{12}^{(2)} & W_{13}^{(2)} & W_{14}^{(2)} \\ W_{21}^{(2)} & W_{22}^{(2)} & W_{23}^{(2)} & W_{24}^{(2)} \\ W_{31}^{(2)} & W_{32}^{(2)} & W_{33}^{(2)} & W_{34}^{(2)} \end{bmatrix} \cdot \max \left\{ 0, \begin{bmatrix} W_{11}^{(1)} & \cdots & W_{1d}^{(1)} \\ \vdots & \ddots & \vdots \\ W_{41}^{(1)} & \cdots & W_{4d}^{(1)} \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix} + \begin{bmatrix} b_1^{(1)} \\ \vdots \\ b_4^{(1)} \end{bmatrix} \right\} + \begin{bmatrix} b_1^{(2)} \\ b_2^{(2)} \\ b_3^{(2)} \end{bmatrix} \right)$$

and finally, the **Combined Forward Pass Expression:**

$$\hat{y} = \phi \left(W^{(3)} \cdot \tanh \left(W^{(2)} \cdot \text{ReLU} \left(W^{(1)}\mathbf{x} + \mathbf{b}^{(1)} \right) + \mathbf{b}^{(2)} \right) + \mathbf{b}^{(3)} \right)$$

(b) **Computing** $\frac{\partial \mathcal{L}}{\partial W_{jk}^{(2)}}$ **and** $\frac{\partial \mathcal{L}}{\partial b_j^{(2)}}$. Let $z_j^{(2)} = \sum_k W_{jk}^{(2)} h_k^{(1)} + b_j^{(2)}$, $h_j^{(2)} = \tanh(z_j^{(2)})$

Gradient wrt $W_{jk}^{(2)}$: $\frac{\partial \mathcal{L}}{\partial W_{jk}^{(2)}} = \frac{\partial \mathcal{L}}{\partial h_j^{(2)}} \cdot \frac{\partial h_j^{(2)}}{\partial z_j^{(2)}} \cdot \frac{\partial z_j^{(2)}}{\partial W_{jk}^{(2)}}$,

where $z_j^{(2)} = \sum_k W_{jk}^{(2)} h_k^{(1)} + b_j^{(2)}$, so

$$\frac{\partial z_j^{(2)}}{\partial W_{jk}^{(2)}} = h_k^{(1)}, \quad \frac{\partial h_j^{(2)}}{\partial z_j^{(2)}} = 1 - \tanh^2(z_j^{(2)}), \quad \delta_j^{(2)} := \frac{\partial \mathcal{L}}{\partial h_j^{(2)}}$$

$$\Rightarrow \frac{\partial \mathcal{L}}{\partial W_{jk}^{(2)}} = \delta_j^{(2)} \cdot (1 - \tanh^2(z_j^{(2)})) \cdot h_k^{(1)}.$$

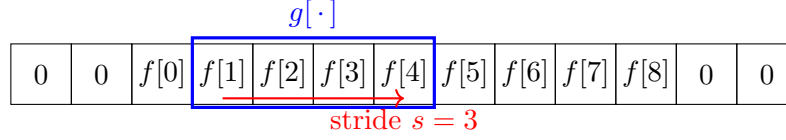
Now using the same logic, we can derive the gradient w.r.t. $b_j^{(2)}$ as: $\Rightarrow \frac{\partial \mathcal{L}}{\partial b_j^{(2)}} = \delta_j^{(2)} \cdot (1 - \tanh^2(z_j^{(2)}))$.

Role of activation derivatives:

- ReLU: $\frac{d}{dz} \text{ReLU}(z) = \begin{cases} 1 & z > 0 \\ 0 & z \leq 0 \end{cases}$; may zero out gradients.

- $\tanh$: $\frac{d}{dz} \tanh(z) = 1 - \tanh^2(z)$; smooth but saturates the small probability values at the clipping extremes.

4. Let the input signal $f = [1, 3, 5, 7, 9, 11, 13, 15, 17]$ is of length $N = 9$ and a filtering kernel $g = [1, -1, 1, -1]$ is of length $m = 4$. A convolution is applied with padding (p) values as zeros and stride $s = 3$ as shown below:



- (a) Formulate the output length L of the filtered signal in terms of the variables N, p, m , and s . Compute the value of L when $N = 9, p = 2, m = 4$, and $s = 1, 2$, and 3 (**3 Marks**)
- (b) Compute the final signal values when f is convolved twice – *i.e.*, the *first* convolution with g and then *second* convolution on the result of the *first* with the $\hat{g} = [-1, 1, -1, 1]$. In another words, computer $((f * g) * \hat{g})$. (**7 Marks**)

Solution and Explanation: (a) The Output Length Formula for 1D Convolution is $L = \left\lfloor \frac{N+2p-m}{s} \right\rfloor + 1$ Where N = input length, p = padding, m = filter size, s = stride. Therefore, when $N = 9, p = 2, m = 4$, and for $s = 1, 2, 3 \Rightarrow$ The value of L for each s :

Case 1: $s = 1, \quad L = \left\lfloor \frac{9+2 \cdot (2)-4}{1} \right\rfloor + 1 = \left\lfloor \frac{9}{1} \right\rfloor + 1 = 9 + 1 = 10$

Case 2: $s = 2, \quad L = \left\lfloor \frac{9+4-4}{2} \right\rfloor + 1 = \left\lfloor \frac{9}{2} \right\rfloor + 1 = 4 + 1 = 5$

Case 3: $s = 3, \quad L = \left\lfloor \frac{9+4-4}{3} \right\rfloor + 1 = \left\lfloor \frac{9}{3} \right\rfloor + 1 = 3 + 1 = 4$

(b) **Double Convolution:** First, pad the input signal $f = [1, 3, 5, 7, 9, 11, 13, 15, 17]$ with $p = 2$ zeros on both sides:

$$f_{\text{padded}} = [0, 0, 1, 3, 5, 7, 9, 11, 13, 15, 17, 0, 0]$$

Convolve f with $g = [1, -1, 1, -1]$ using stride $s = 3$:

$$\begin{aligned} [0, 0, 1, 3] \cdot g &= 0 + 0 + 1 - 3 = -2 \\ [3, 5, 7, 9] \cdot g &= 3 - 5 + 7 - 9 = -4 \\ [9, 11, 13, 15] \cdot g &= 9 - 11 + 13 - 15 = -4 \\ [15, 17, 0, 0] \cdot g &= 15 - 17 + 0 + 0 = -2 \end{aligned}$$

$$f * g = [-2, -4, -4, -2]$$

Now, convolve the result with $\hat{g} = [-1, 1, -1, 1]$ using the same settings:

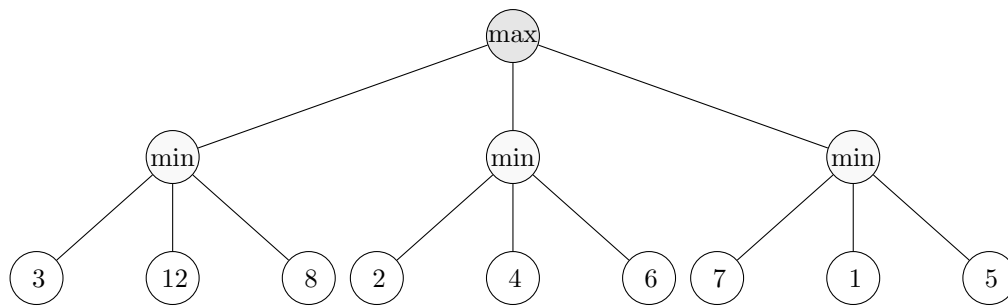
$$[-2, -4, -4, -2] \rightarrow [0, 0, -2, -4, -4, -2, 0, 0]$$

$$\begin{aligned} [0, 0, -2, -4] \cdot \hat{g} &= 0 + 0 + 2 - 4 = -2 \\ [-4, -4, -2, 0] \cdot \hat{g} &= 4 - 4 + 2 + 0 = 2 \end{aligned}$$

$$((f * g) * \hat{g}) = [-2, 2]$$

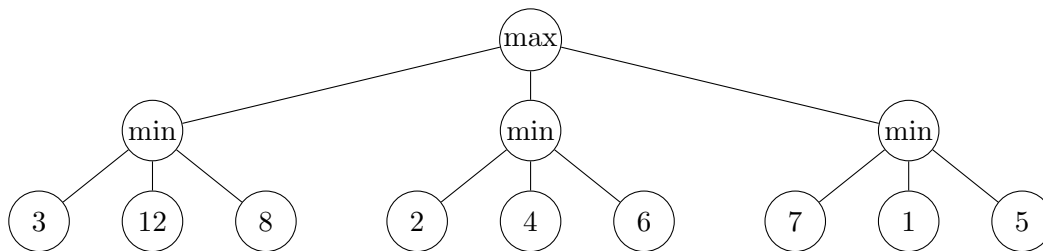
5. A start-up *HelloChess* is designing a two-player, zero-sum board game. The game always ends after three moves (**Note:** one move by either player is called a *ply*). The team

assigned a utility value to every ending position and drew the full 3-ply game tree shown below. The search algorithm examines each set of child nodes from left to right.



- Compute the minimax value of every internal node and state ****which move (Max's action)**** is chosen at the root. Show your working clearly. **(4 marks)**
- $\alpha-\beta$ pruning. Traverse the **same** tree using $\alpha-\beta$ search with the *given* child order. At each node indicate (i) its current (α, β) window when first entered, and (ii) any branches that are pruned (cross them out in the figure). How many terminal nodes are *not* evaluated thanks to pruning? Draw the diagram of the tree after every pruning. **(4 marks)**
- Suppose the leaf value '12' is changed to an unknown value $k \in \mathbb{Z}$. Determine the range of k for which Max's optimal action at the root would *change*. Justify your answer carefully. **(2 marks)**

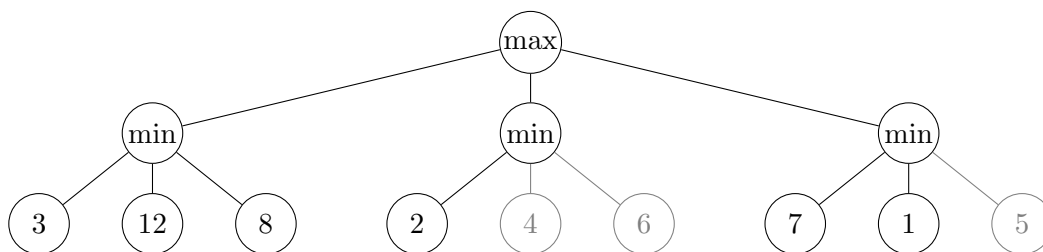
Solution and Explanation: (a) Minimax Evaluation



Minimax values:

$\text{min1} = \min(3, 12, 8) = 3$, $\text{min2} = \min(2, 4, 6) = 2$, $\text{min3} = \min(7, 1, 5) = 1$
 $\Rightarrow \text{max} = \max(3, 2, 1) = 3$ (left move chosen)

(b) Alpha-Beta Pruned Tree



1. Root (max): $\alpha = -\infty$, $\beta = +\infty$

2. **Visit min1** ($\alpha = -\infty, \beta = +\infty$): Visit 3 $\Rightarrow \beta = 3$, Visit 12 $\Rightarrow \beta = \min(3, 12) = 3$, Visit 8 $\Rightarrow \beta = \min(3, 8) = 3 \Rightarrow$ Return 3, update $\alpha = 3$
3. **Visit min2** ($\alpha = 3, \beta = +\infty$): Visit 2 $\Rightarrow \beta = 2$, Now $\beta = 2 < \alpha = 3 \Rightarrow$ **Prune 4, 6** $\Rightarrow$ Return 2
4. **Visit min3** ($\alpha = 3, \beta = +\infty$): Visit 7 $\Rightarrow \beta = 7$, Visit 1 $\Rightarrow \beta = \min(7, 1) = 1$, Now $\beta = 1 < \alpha = 3 \Rightarrow$ **Prune 5** $\Rightarrow$ Return 1
- Note:** It is also acceptable if the pruning is shown additionally on the Tree itself.

Pruning Summary– Pruned nodes: 4, 6, 5 and Total pruned leaf nodes: **3**

(c) **Root Decision Change Condition** If k replaces leaf value 12, then $\min1 = \min(3, k, 8)$. To change root value from 3 to another, need: $\min1 < 3 \rightarrow k < 3$. Therefore, the Root decision changes if and only if $k < 3$.

6. The below tabular data shows that an e-commerce firm recorded purchase behavior of eight customers, each represented by two normalized features ($x_1 = \text{frequency}, x_2 = \text{average basket value}$). The marketing team wishes to form $K = 2$ clusters.

Customer	x_1	x_2	ID
1	1.0	1.2	P_1
2	1.5	1.8	P_2
3	2.0	1.0	P_3
4	2.2	1.2	P_4
5	3.8	3.0	P_5
6	3.2	2.5	P_6
7	4.0	2.8	P_7
8	3.5	3.2	P_8

Point	$d(C_1^{(0)})$	$d(C_2^{(0)})$	Assigned Cluster
P_1	0.00	3.40	C_1
P_2	0.71	2.62	C_1
P_3	1.00	2.68	C_1
P_4	1.20	2.48	C_1
P_5	3.22	0.28	C_2
P_6	2.50	0.85	C_2
P_7	3.40	0.00	C_2
P_8	3.44	0.51	C_2

- (a) Formulate a heuristic to ‘choose the two most distant points’ for setting the initial centroids $C_1^{(0)}$ and $C_2^{(0)}$ and compute the distance between them. **(2 marks)**
- (b) **Draw** and **Calculate** all the distances and mean calculations at every iteration of your clustering steps name the centroids. Name the recomputed centroids $C_1^{(i)}, C_2^{(i)}$ at the i^{th} iteration. **(5 marks)**
- (c) Check the convergence of the algorithm by calculating the within-cluster sum of squared errors (SSE). Justify your answer. **(3 marks)**

Solution and Explanation: Given: 8 customers with normalized features (x_1, x_2) . Cluster into $K = 2$ groups.

Table 1: SSE Derivation Matrix for Final Clusters

Point ID	Cluster	x_1	x_2	Centroid x_1	Centroid x_2	Squared Error
P ₁	C ₁	1.0	1.2	1.675	1.300	0.4656
P ₂	C ₁	1.5	1.8	1.675	1.300	0.2806
P ₃	C ₁	2.0	1.0	1.675	1.300	0.1956
P ₄	C ₁	2.2	1.2	1.675	1.300	0.2856
P ₅	C ₂	3.8	3.0	3.625	2.875	0.0462
P ₆	C ₂	3.2	2.5	3.625	2.875	0.2962
P ₇	C ₂	4.0	2.8	3.625	2.875	0.1531
P ₈	C ₂	3.5	3.2	3.625	2.875	0.1396
Total SSE						1.8625

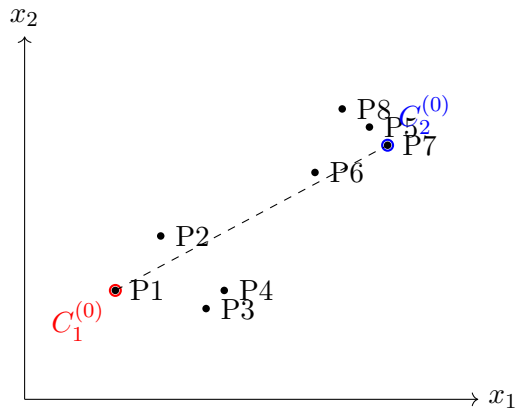
(a) Heuristic: Choose Two Most Distant Points

We use Euclidean distance to select the two most distant points:

$$d = \sqrt{(x_1^{(i)} - x_1^{(j)})^2 + (x_2^{(i)} - x_2^{(j)})^2}$$

Max distance is between $P_1(1.0, 1.2)$ and $P_7(4.0, 2.8)$:

$$d = \sqrt{(4.0 - 1.0)^2 + (2.8 - 1.2)^2} = \sqrt{9 + 2.56} = \sqrt{11.56} \approx \boxed{3.4}$$

**(b) Iterative Assignment and Centroid Recalculation**

Iteration 0:

$$C_1^{(0)} = (1.0, 1.2), \quad C_2^{(0)} = (4.0, 2.8)$$

Clusters:

- C_1 : $\{P_1, P_2, P_3, P_4\}$
- C_2 : $\{P_5, P_6, P_7, P_8\}$

Iteration 1: Compute new means:

$$C_1^{(1)} = \left(\frac{6.7}{4}, \frac{5.2}{4} \right) = (1.68, 1.3), \quad C_2^{(1)} = \left(\frac{14.5}{4}, \frac{11.5}{4} \right) = (3.625, 2.875)$$

Clusters unchanged $\Rightarrow$ Converged. Consider the distance and cluster matrices above

(c) Convergence Check: SSE

$$SSE = \sum_{i=1}^n \|x_i - C_k\|^2 = \boxed{1.8625}$$

Interpretation: Compact clusters, minimal intra-cluster variance. Convergence achieved.